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COURSE : DATA ANALYTICS AND DATA SCIENCE

SECTION A : DATA FUNDAMENTALS

1. Database → is an organized collection of data that is stored that is stored electronically in a computer system that can be managed, accessed and updated easily.
   - The database is essential in the morden world as it helps manage influx or massive amounts of data and decision-making.
   - Examples of where they are used: Healthcare Systems and Call Centre Systems.

2. Data Warehouse → A System that houses / stores large amount of databases.
   - Data Warehouse consists of multiple databases stored centrally; while a database include 1 database
   - * Companies use data Warehouse as it simplifies the data within it and easily acisible and easy to make business decisions based on the outcomes.

3. Data Lake → A system that stores unprocessed data warehouses.
   - Data warehouse is multiple databases that are processed and data cleaned, while data lake is raw, and unorganized warehouses.
   - Banking businesses would prefer it as it would include the transaction loggs, Fraud department

4. Data Mart → is a smaller, more focused on data ware house that stored or is designed for a specific business function or department.

- Data Mart is derived from an existing data warehouse and [redacted] example would be Finance department that would typically analyze the budget and expenses within a department.

5. Structured Data → Data that is organized and formatted.
   eg: Customer_ID

   Semi - Structured → Data that is partially organized
   eg: Emails.

   Unstructured Data → Data that is not organized and not formatted
   eg: Social Media posts.

6. Data Modelling → It's a process of creating a visual and logical representation of data.

- It is important because it makes everyone understand how data is organized in databases, and how to build and maintain them.
- Two types of data modelling
  1. Logical data model.
  2. Physical data model

7. Data Mining → is the process of analyzing large amounts of data to discover the patterns and useful information that helps in making business decisions.

- Data mining is used to understand customer behaviour and improve the business decision making that ~~th~~ can help in detecting fraud and any risks that can affect the business.
- Two ~~realife~~ real-world examples of data mining
  1. Telecommunications
  2. Health care

8. Database Management System (DBMS)

→ DBMS is a software that ~~allows~~ allows users to create, store, organize, retrieve and manage data in a structured way.

- DBMS Platforms
  1. MySQL → Stores data in tables with rows and columns using SQL
  2. PostgreSQL → is an opensource DBMS with strong support for complex queries and data integrity
  3. Oracle Database → Handles very large databases
  4. MongoDB → Handles unstructured data well.

* MySQL usecase
  → Used for e-commerce business eg (managing customers, products)
* PostgreSQL usecase
  → Best for data analytics or financial applications where accuracy and complex queries are critical.
* Oracle Database
  → Suitable for large organizations e: Banks
* MongoDB
  → Works best for applications that changes regularly.

9. ETL → it's an intergrated process used to collect data from diffrent types of sources, clean the data and prepare it and store it in a central system for analysis

- Extract → Data is collected from multiple sources such as database.
- Transform → Data is cleaned and converted into a suitable format
- Load → The transformed data is loaded into a targated system like data warehouse.

10. Common data storage formats used in data analytics

1. CSV (Comma-separated values)
   - Plaintext that is stored in rows and ~~lea~~ columns and separated by commas
   - ∴ used in small datasets.
2. JSON (Javascript Object Notation)
   - Semi-structured data that is stored in key-value pairs.
   - ∴ used in web applications and API's.
3. XML (eXtensible Markup language)
   - Semi-structured that uses tags to define data structure
4. Parquet
   - Column storage format, used in big data analytics platforms

11. Artificial Intelligence (AI)

→ is a technology that enables machines to perform tasks that are typically requiring human intelligence, such as recognizing pattern, making decisions and solving complex problems

* Narrow AI → is AI designed to perform a specific task or is limited to a few tasks

eg: Google Assistant.

* General AI → AI with the ability to understand, learn and apply intelligence across a wide range of tasks, similar to a human.

es: ChatGPT

12. Machine Learning (ML)

→ Its a branch in AI that enables computers to learn from data and improve their performance without being explicitly programmed

- ML is different to traditional programming because you provide data and expected outcomes

* Three types of machine learning

1. Supervised Learning

→ The model is trained using labeled data

2. Unsupervised learning

→ The model works with unlabeled data

3. ~~Reinforced~~ Reinforcement learning

→ The model learns by interacting with the environment and receiving awards

13. Deep Learning → is a subnet of machine-learning based on Artificial ~~Neural~~ neural networks with multiple layers

14. LLM → They are AI models trained on massive datasets using transformer architecture to understand, summarize, generate and predict content, including code, text and images

3 real-world applications of LLM

1. Healthcare documentation and diagnosis
2. Financial fraud detection and risk assessment
3. E-commerce personalization and customer support

16. Generative AI → The broader category of AI that creates new, original content by leaining from existing data

* At higher level it works as a sophisticated pattern mimic

3 tools that uses it.

1. ElevenLabs
   - This tools focuses on voice synthesis and music creation
2. Midjourney
   - This tool is for image and video generation
3. Gemini (google)
   - This tool that can process and generate text, images and videos

17. Training dataset → is the initial dataset used to teach AI models to recognize pattern, make predictions or perform tasks

- The training data quality is critical as it ensures the data accuracy and reliability

* If training data is wrong or biased it can cause physical danger, legal or financial liabilities.

18. Computer Vision (CV) is a subfield of AI that trains machines to process, analyze and interpret visual data from the world

* Machines do not see images the way humans do: instead they interpret visual data through a structured, numerical and algorithmic approach

- Real world use cases of computer vision
  1. Medical imaging and diagnosis
     → Computer vision algorithms analyze medical scans e.g X-rays
  2. Retail Analytics
     → Camera tracks customers and products they pick

19. API – is a set of rules and protocols that allows different software applications to communicate with each other

- API's are the backbone of modern connectivity

20 API Key → is a unique identifier used to authenticate a user or application making a request to an API

- Purpose : It acts as a digital "password" to track and control how the API is being used to prevent abuse. For paid services

21 REST API → is an architectural style for providing interoperability between computer systems on the internet.

- REST stands for : REpresentational State transfer
- Main HTTP methods

GET : Retrieves data from a server

POST : Sends new data to a server to create a resource

PUT : Updates an existing resource on server entirely

DELETE : Removes a specific resource from the serve

22. JSON → is a lightweight, text-based format for storing and transporting data, It is easy for humans to understand as its structure is simple

eg JSON

```
{ "name" : "Alex"
  "role" : "Data Analyst"
}
```

23 Prompt Engineering → is the process of crafting refining and optimizing inputs to guide AI models towards generating the most accurate, relevant and high-quality outputs

- Importance: since AI models respond based on the context and context and instructions provided.

Two ~~emple~~ examples of poorly written prompts.

1. "Write a story"
2. "Analyze this data"

24 MAP Function → is a higher-order function that applies a specific transformation to every item in a collection.

25 Cloud Computing → is the on-demand delivery of computing services, including servers, storage, databases and software over the internet

Three main cloud services models

1. IaaS (Infrastructure as a service)
   - Provides fundamental computing resources.
2. PaaS (Platform as a service)
   - provides a framework for developers to build and deploy apps without worrying about underlying hardware or OS management
3. SaaS (Software as a service)
   - Delivers ready to use software applications over the internet

26. Version Control System → is a tool that tracks and manages changes to software code ~~and~~ or data projects over time.

- Git → is a tool that allows teams to work on the same files simultaneously without overwriting each other's work.

1. Commit: A snapshot of your project at a specific point of time
2. Branch: A parallel version of your project. It allows ~~you~~ to work on a new feature or experiment in isolation without affecting the "main" code.

27 Jupyter notebook → is an open-source web application that allows you to create and share documents containing live code, equations visualizations and narrative text.

- its popular because it allows users to run "cells" of code individually and see results immediately beneath the code.
  - its easy to share HTML files for collaborations.

28. Python → is a high-level interpreted, general-purpose programming language known for its readability and simplicity

Four common libraries

1 Pandas → used for data manipulation and analysis
2 Numpy → The foundation for ~~specific~~ scientific computing: it handles large, multi-dimensional arrays and complex mathematical functions
3 Matplotlib → A plotting library used for creating visualizations
4 Scikit-learn → The "go-to" library for machine learning, providing tools for data mining, classification and regression.

29. Data Privacy → Refers to the right of individuals to control how their personal information is collected, used and shared

GDPR - is one of the world's toughest privacy and security laws it sets strict rules on how organizations handle personal data and grants individuals rights ~~like~~.

30. Data Ethics → refers to the branch of ethics that evaluates the moral implications of how data is collected, shared and used, focusing on fairness, transparency and accountability

Examples

- Surveillance without consent
  - using facial recognition in public spaces without notifying citizens

- Discriminatory Algorithms – A hiring tool that filters out candidates based gender or race due to historical bias in the training data
- Data brokering – Selling user health or location data to third party

31. Data Governance → Is the process of collection of processes, roles policies and standards that ensure an organization's data is accurate, secure and used efficiently

Key Components

1. Data quality → Ensure data is accurate, complete and reliable
2. Data Stewardship → Assigning individuals responsible for managing and protecting specific data sets
3. Data Security and Privacy → Establishing rules for who can access data and how it is protected from breaches
4. Compliance → Ensuring the organization follows legal regulations like GDPR

32. BIAS in AI → It occurs when an algorithm produces results that are systematically prejudiced due to erroneous assumptions in the machine learning process

33. Key difference between a data analyst, Data Scientist, Data engineer and AI Engineer.

Role	Primary goal	Key skills
Data Analys	interprets data to provide actionable Insights	SQL, Data visualization
Data Scientist	Uses advanced math/coding to Predict Future trends	Machine learning, Python, R

Role	Primary goal	Key skills
Data Engineer	Builds system to transport and store data	machine learning, ETL, coding
AI Engineer	Develops and deploys functional AI model/applications	Deep learning, Software Dev

34 Business Intelligence → Compromises the strategies and technologies used by enterprises for data analysis for business information

* Two popular BI tools
  1. power BI
     → They help visualization and ~~tela~~ real-time monitoring
  2. Tableau

35 Machine learning topics, how they differ amongst other AI tools

36. Machine learning and deep learning, I used more of youtube videos and chatgpt to help me understand

37. I want to grow more into the field as the current tech field has limitations, I want to understand data and analyze it more now.

References: